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A comparison of different modelling solutions for studying grapevine phenology under present and future climate scenarios

Gianni Fila[a,*], Massimo Gardiman[a], Paolo Belvini[b], Franco Meggio[c], Andrea Pitacco[c]

[a] Consiglio per la Ricerca e la sperimentazione in Agricoltura – Centro di Ricerca per la Viticoltura, I-31015 Conegliano, TV, Italy
[b] Centro per l'Educazione la Cooperazione e l'Assistenza Tecnica (CECAT), I-31033 Castelfranco Veneto, TV, Italy
[c] Università di Padova, Dipartimento di Agronomia Animali Alimenti Risorse Naturali e Ambiente – DAFNAE, Agripolis, I-35020 Legnaro, PD, Italy

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ABSTRACT

This paper evaluates a set of candidate phenology models for simulating grapevine development under present conditions and future climate change scenarios. The models were developed on a combination of working hypotheses, designed to extend their predictive ability to multiple phenological events, i.e. budburst, flowering and veraison.

The aspects considered in the modelling analysis were: (i) effect of dormancy description, by comparison of Chilling and Forcing (CF) with Forcing (F) models; (ii) degree of phenophase-dependent specificity after endodormancy break; (iii) adopting a linear or curvilinear function for modelling temperature dependency of development rate; (iv) performance under climate change scenarios.

We also analyzed to what extent model behavior under present and future climate was affected by factors other than model structure, focusing on the variability of the dataset used to calibrate the models. To this purpose two parallel series of models were evaluated: one calibrated on a historical phenological dataset, and the other on the same dataset extended with budburst observations obtained under controlled condition simulating shorter winters, as expected from climate change predictions.

Estimation accuracy of the models was evaluated on independent data. A further evaluation considered parameterization consistency with experimental physiological data, and model behavior under future climate change scenarios, to assess to what extent future predictions are affected by model structure and complexity.

Under current climatic conditions the models showed a comparable accuracy, only slightly higher when growth-room data were used for calibration.

The similar performance was obtained despite wide discrepancies in parameterizations between the calibration types. In CF models, which describe dormancy dynamics, chilling requirements were highly overestimated in field-calibrated models, and different temperature requirements were likewise found for post-dormancy development.

The parameterization differences had a clear impact on a demonstrational scenario analysis, where models were driven by synthetic weather data derived from downscaled HadCM3 predictions for the SRES-A2 scenario.

Here all the selected models predicted a forward shift of phenology. The type of calibration affected the response more than the model structure or complexity, the effect of which only became detectable at the remotest projections in the future. The effect of calibration was more evident for budburst than for flowering and veraison. Field-calibrated models underpredicted the shift with respect to their counterparts, due to a delayed onset of forcing accumulation, which postponed budburst and subsequent phases, counterbalancing the effect of higher spring temperatures.

It is concluded that for present time applications all the evaluated models have comparable accuracy. Also for climate change impact analysis the variability of the calibration dataset is a more crucial factor than model structure or complexity. The simpler forcing models can therefore be recommended for both present conditions as well as scenario-based applications.

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* Corresponding author. Tel.: +39 438 456778; fax: +39 438 450773.
E-mail address: gianni.fila@entecra.it (G. Fila).

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1. Introduction

Grapevine is arguably one of the crops most challenged by climate change (Goode, 2012; Jones and Webb, 2010). In traditional wine-growing regions, century-long trial and error processes have optimized the combination of cultivar/rootstock, soil, climate and management, so that the local environment suits the grapevine thermal and moisture requirements, yielding high-quality grapes (Jones et al., 2004; Schultz and Jones, 2010). When this delicate equilibrium is achieved, berries ripen after the hottest period of the year, with moderate temperatures and higher daily diurnal temperature ranges suitable for preserving acidity, aromas and color. Occasional forward shifts of development timing in warmer years cause berries to ripen at higher temperatures, with a detrimental impact on quality (Jackson and Lombard, 1993; Mori et al., 2007).

Advancements in grapevine phenology are reported to have occurred over the last decades due to climate change (Jones and Davis, 2000; Nemani et al., 2001; Wolfe et al., 2005; Jones et al., 2005; Menzel, 2005; Petrie and Sadras, 2008; Ramos et al., 2008; Bock et al., 2011; Duchêne and Schneider, 2005), with detectable effects on berry composition and wine quality (Jones et al., 2005; Duchêne and Schneider, 2005; Laget et al., 2008; Mira de Orduña, 2010; Neethling et al., 2012). According to the IPCC Fourth Assessment Report (IPCC, 2007), the warming trend is expected to continue in the future, so the ripening phase will be characterized by higher temperatures with increasing frequency, worsening berry quality. This is likely to put established typical varieties at risk in many areas, if not the feasibility of grapevine growing itself.

In this context, understanding how suitability for viticulture is changing is of primary interest for at least two reasons: to allow the development of adaptation strategies in traditional wine-growing regions, and to explore cultivation potential in previously cool ones, where climate change is moving towards favorable thermal conditions (Jones, 2012).

Whatever the objective, reliable site-specific quantitative assessments are necessary to implement practical local guidelines for the adaptation of viticulture.

This has led to renewed interest in phenology models, which are among the main tools currently available to predict grapevine development under future climate change scenarios. A number of model-assisted studies concerning climate change impact on grapevine have appeared in recent years (Webb et al., 2007; Duchêne et al., 2010; Caffarra and Eccel, 2011). These and other modelling efforts all follow a typical approach, with three sequential steps: (i) set up a phenology model driven by climate variables, notably temperature, and calibrate it on historical data; (ii) generate climate projections; (iii) run the model on these projections and analyze the results.

In order to properly characterize the suitability of a given grapevine cultivar to a given environment, a full assessment of the development cycle is necessary to understand whether berries can reach maturation under the local climate, hence, models predicting multiple phenological events are required. Extending the predictive ability is a delicate issue however, since it inevitably raises model complexity, and thus the risk of overfitting effects, which occur when a model describes the random error in experimental data rather than the underlying relationships.

The most popular models used to describe grapevine phenology belong to the forcing (F) type, which consists in the accumulation of forcing units starting from a fixed date, most often January 1st. Various forcing-units definitions exist, calculated on a daily or hourly basis. The most widely used are Growing Degree Days (Wang, 1960; Winkler et al., 1974; Bonhomme, 2000), which are defined as the difference between the mean daily temperature (T_{mean}) and a base temperature (T_b): GDD = ($T_{mean} - T_b$), for $T_{mean} > T_b$.

Another broad family of models is represented by the Chilling and Forcing (CF) type, which include the description of the dormancy phase (e.g. Richardson et al., 1974; Cannell and Smith, 1983; Chuine, 2000; Linkosalo et al., 2008). In the "sequential" sub-family of CF models, which is the most widely adopted, the estimated dormancy exit date marks the starting point of forcing units accumulation.

Having a wider explanatory basis, as a result of their sensitivity to the winter thermal time course, they have greater potential than forcing models to capture year-to-year variability. However, whether F or CF models should be preferred to model grapevine phenology is still a debated issue, since direct comparisons have so far not given unanimous results. CF models have been found to provide comparable (García de Cortázar-Atauri et al., 2009) or better estimates for budburst (Hlaszny et al., 2011) and for the subsequent phases of flowering and veraison (Caffarra and Eccel, 2010) with respect to the former. In other studies they showed comparable or poorer performances (Parker et al., 2011).

The reason why CF does not easily outperform F models can be twofold. On the one hand, CF models are more prone to overfitting due to their higher complexity, so they tend to be less robust when tested on independent data. On the other hand, under present climatic conditions in most wine-growing regions, grapevine chilling requirements are fulfilled before January, long before forcing units accumulation begins, so the precise determination of dormancy break date is not critical (Vitasse et al., 2011; Linkosalo et al., 2008; Thompson and Clark, 2008).

Indeed, in terms of simplicity and robustness forcing models still appear to be a better choice under present climatic conditions. However this may not be true for climate change studies, when winters are expected to become shorter and milder, increasing the probability that chilling requirements will be slowly or even insufficiently fulfilled. The opposite effect, i.e. accelerating chilling fulfillment, is also possible in cold regions, when the warming trend takes winter temperatures near the optimal chilling temperature, especially when this is above 0 °C (Linkosalo et al., 2000). CF models are theoretically capable of discriminating between these two situations, provided they implement consistent temperature-dependent functions of the chilling fulfillment rate.

With models describing more than one phenological phase, complexity is strongly affected by the specificity level applied to describe each phase. This concerns whether the post-dormancy development can be estimated using the same thermal requirement, or specific conditions must be adopted for each phenophase (e.g. Moncur et al., 1989; Duchêne et al., 2010; Oliveira, 1998).

The level of model complexity also depends strongly on the mathematical relationship adopted to describe how the development rate varies with the driving variables, most notably temperature. The relationship can be described by a linear dependency on temperature, as in the popular Growing Degree Days (GDD) approach, or by curvilinear functions, where the effects on development rate change with temperature (e.g. Chuine, 2000).

In this study we attempted to refine phenology modelling from budburst to veraison addressing the issues outlined above, with the ultimate goal of improving grapevine modelling for current research and practical applications.

Harvest was not included in the modelling task, because the criteria used for individuating harvest time were seldom homogeneous, having changed over the years following technical developments.

We evaluated a set of modelling solutions built around existing modelling concepts, focusing on the following aspects: (i) assess the importance of including dormancy description in models; (ii) degree of model complexity: does a higher parameterization improve model performance? (iii) linear vs. curvilinear model of

temperature dependency of post-dormancy development; (iv) analyze how the above-mentioned aspects affect model performance under scenario conditions.

Beyond the intrinsic mathematical structure, model parameterization is another key factor affecting model performance. Models are typically optimized against observational phenological time series, most often covering present time conditions. A question which to our knowledge has never been given full attention yet, is whether present-based parameterizations are adequate to simulate plant phenology under future conditions. In case datasets taken under different climatic conditions were available for model calibration, do the derived parameterizations react differently under present or future conditions? And if this is so, how to choose the most adequate one? We addressed this issue in our analysis, by studying the responses of differently parameterized models under both present and future scenario conditions.

2. Materials and methods

2.1. Analysis layout

The evaluated models are described in Table 1. They were designed to gain a better understanding of the issues outlined below.

2.1.1. CF vs. F models

The proposed CF and F models drew their basic ideas from the 'UniChill' and "UniForc" models, developed by Chuine (2000). These models were originally conceived to estimate budburst date, but they have already been used to estimate flowering and veraison (Parker et al., 2011).

In CF models, the duration of the endodormancy period is estimated by accumulating daily chilling units (c.u.) from September 1st of the year preceding the phenological observations of budburst, flowering and veraison, until a critical sum is reached (C_{crit}), marking the end of dormancy. From this stage they accumulate forcing units (f.u.) up to another critical sum triggering budburst (Sbb). Forcing units accumulation then restarts towards two further sums triggering flowering (Sfl) and veraison (Sve). This model belongs to the "sequential" sub-family of CF models, where dormancy and forcing are well-separated phases, one immediately after the other.

The F models simulate only thermal accumulation from a fixed date, by which dormancy is assumed to have already ended. Forcing units are calculated up to the critical sums Sbb, Sfl and Sve, which have the same meaning as in CF models. The starting date of forcing accumulation (t_0) was optimized against experimental data, since this was found to significantly improve models goodness of fit in previous studies (Williams et al., 1985; Duchêne et al., 2010; Nendel, 2010; Parker et al., 2011).

2.1.2. Development rate dependency on temperature

Daily rates of chilling and forcing were modelled using a beta-type function (Yan and Hunt, 1999; Amaducci et al., 2008), which is fully described by three cardinal temperatures:

$$\begin{cases} r = \left(\frac{T_{\max} - T}{T_{\max} - T_{opt}}\right) \cdot \left(\frac{T - T_{\min}}{T_{opt} - T_{\min}}\right)^{\left(\frac{T_{opt} - T_{\min}}{T_{\max} - T_{opt}}\right)} & T > T_{\min}; \quad T < T_{max} \\ r = 0 & T \le T_{\min}; \quad T \ge T_{\max} \end{cases} \quad (1)$$

where r is the daily chilling or forcing unit; T is the daily mean temperature; T_{min} is the minimum temperature, below which chilling (or forcing) rate = 0. T_{max} = maximum temperature, above which chilling (or forcing) rate = 0. T_{opt} = optimal development temperature, at which the highest chilling (or forcing) rate occurs (=1).

In the original UniChill model forcing units are computed by means of a two-parameter logistic curve, which increases reaching a plateau. Adoption of a beta-type curve allows to take into account temperatures with inhibiting action on development (García de Cortázar-Atauri et al., 2010). An additional advantage of the beta-type curve is that the cardinal temperatures T_{min}, T_{opt} and T_{max} have a physiological meaning, which allows model consistency to be checked against experimental data.

In order to limit the total number of parameters, T_{min} and T_{max} were set at arbitrarily assigned fixed values (García de Cortázar-Atauri et al., 2010). T_{min} was set at 0 °C, assuming that no growth occurs below this, while for T_{max} the following discrete values were evaluated: 30, 35, 40 and 45 °C. All models were optimized and validated using each of these values, and the T_{max} which gave the best validation output was finally kept. Hence, all the forcing functions differed only for the optimal temperature. Validity limits for the chilling functions were instead optimized against observations.

The beta-function based F models were compared with the classic GDD model, as this still represents one of the most widely-used approaches in grapevine phenology modelling.

Beta- and GDD-based F models were labelled F_{beta} and F_{GDD} models respectively.

2.1.3. Phenophase-specificity of forcing accumulation

Three different solutions were formulated for post-dormancy modelling of phenophases. For the sake of simplicity, the same labelling was adopted for T_{opt}, which refers to the beta function, and T_b used for GDD calculation, as well as for the derived threshold forcing units summation, but it must be kept in mind that they have different meanings in the two modelling approaches.

1. Forcing units (f.u.) were calculated with the same T_{opt}/T_b for budburst, flowering and veraison.
2. f.u. were calculated with one T_{opt} (or T_b) for budburst, and with a second one accounting for both flowering and veraison.
3. f.u. were calculated with a specific T_{opt} (or T_b) for each of the three phases.

Irrespective of how many T_{opt} or T_b were adopted, there were always three forcing threshold sums (S), one for each phenophase (S_{bb}, S_{fl}, S_{ve}).

Following the above criteria, all CF and F models were then furthered labelled with the total number of parameters (Table 1).

2.2. Phenological observations

2.2.1. Field observations

Historical phenological records were available for the 'Chardonnay' cultivar in two localities in the Veneto region, North-Eastern Italy:

- *Susegana* (45.85N, 12.26E). Budburst, flowering and veraison dates had been recorded on the ampelographic collection of the Viticulture Research Centre of Conegliano (CRA-VIT). Observations covered the period 1969–2011, with two missing years (2003 and 2007).
- *Istrana* (45.69N, 12.09E). Data were from a CECAT (Centre for Education, Cooperation and Technical Assistance) experimental vineyard. Observations had been made between 1986 and 2011 by one of the authors (P. Belvini), making the dataset highly homogeneous.

Table 1
Description of evaluated models.

Model	Temperature response function		Chilling temperatures	Forcing temperatures	Chilling/forcing critical sums	Start day of forcing accumulation
	Chilling	Forcing				
CF8	Beta	Beta	T_{min}, T_{opt}, T_{max}	$T_{bb\text{-}ve}$	C_{crit}, S_{bb}, S_{fl}, S_{ve}	
CF9	Beta	Beta	T_{min}, T_{opt}, T_{max}	T_{bb}, $T_{fl\text{-}ve}$	C_{crit}, S_{bb}, S_{fl}, S_{ve}	
CF10	Beta	Beta	T_{min}, T_{opt}, T_{max}	T_{bb}, T_{fl}, T_{ve}	C_{crit}, S_{bb}, S_{fl}, S_{ve}	
$F_{beta}5$		Beta		$T_{bb\text{-}ve}$	S_{bb}, S_{fl}, S_{ve}	t_0
$F_{beta}6$		Beta		T_{bb}, $T_{fl\text{-}ve}$	S_{bb}, S_{fl}, S_{ve}	t_0
$F_{beta}7$		Beta		T_{bb}, T_{fl}, T_{ve}	S_{bb}, S_{fl}, S_{ve}	t_0
$F_{GDD}5$		Linear		$T_{bb\text{-}ve}$	S_{bb}, S_{fl}, S_{ve}	t_0
$F_{GDD}6$		Linear		T_{bb}, $T_{fl\text{-}ve}$	S_{bb}, S_{fl}, S_{ve}	t_0
$F_{GDD}7$		Linear		T_{bb}, T_{fl}, T_{ve}	S_{bb}, S_{fl}, S_{ve}	t_0

Legend

Parameter	Units	Explanation
T_{min}, T_{opt}, T_{max}	°C	Minimum, optimal and maximum chilling-effective temperatures
$T_{bb\text{-}ve}$	°C	Optimal (beta function) or base temperature (GDD), for budburst, flowering and veraison
T_{bb}	°C	Optimal (beta function) or base temperature (GDD) for budburst
T_{fl}	°C	Optimal (beta function) or base temperature (GDD) for flowering
T_{ve}	°C	Optimal (beta function) or base temperature (GDD) for veraison
$T_{fl\text{-}ve}$	°C	Optimal (beta function) or base temperature (GDD) for flowering and veraison
C_{crit}	Days	Critical sum of chilling-units marking the end of endodormancy in CF models
S_{bb}, S_{fl}, S_{ve}	Days (F_{beta}); °C*day (F_{GDD})	Critical sums of forcing units (calculated with beta-function or GDD) triggering budburst, flowering and veraison
t_0	DOY	Starting day of forcing accumulation in F models

Budburst, flowering and veraison were recorded in correspondence to the BBCH phases 09, 65 and 81 respectively (Lorenz et al., 1994). Both study sites were equipped with automatized weather stations measuring air temperature and precipitation.

2.2.2. Growth-room experiments

Dormant one-bud cuttings from the 'Chardonnay' cultivar were repeatedly sampled from the Susegana ampelographic collection during two consecutive autumn/winter seasons between 2010 and 2012. The aim was to collect samples with a high variability in chilling status (Table 2).

Sampling started when at least seven days were recorded with daily mean temperature below 10 °C (Dokoozlian, 1999). This condition was met at the end of November in both seasons.

From this moment onwards, cuttings were excised at approximately 15–20 day intervals. In the 2010/2011 campaign sampling ended in mid-April 2011, while in 2011/2012 it ended in mid-February 2012. Overall, during the two campaigns explants were taken 20 times.

Each sample ($n = 30$) was placed in a growth room under semi-controlled conditions, where diurnal temperature ranged between 18.0 ± 2.5 and 25 ± 2.5 °C. Budburst was recorded when 50% of the buds reached stage 09 of the BBCH scale for grapevine.

Table 2
Growth-room experiments were performed to obtain budburst observations to extend the variability of the calibration dataset. In order to have material with a wide range of chilling exposure, cuttings were excised at various dates during Autumn–Winter, listed in the table.

	2010/2011	2011/2012
1	28 Oct	31 Oct
2	12 Nov	4 Nov
3	24 Dec	11 Nov
4	14 Jan	24 Nov
5	24 Jan	1 Dec
6	2 Feb	28 Dec
7	11 Feb	12 Jan
8	22 Feb	2 Feb
9	8 Mar	14 Feb
10	25 Mar	
11	12 Apr	

2.3. Model fitting and validation

All models were calibrated on the Istrana dataset. Validation was performed by running the fitted models on the Susegana temperature data, comparing the obtained phenological response with those observed in this locality, and deriving performance statistics. A second calibration cycle was run on an extended dataset, compiled adding growth-room (GR) data to the Istrana dataset. The resulting extended dataset was named Istrana*. Accordingly, models calibrated on the Istrana* dataset were also labelled with an asterisk, e.g. CF10* (=CF10 model calibrated on Istrana + GR data), and CF10 (=CF10 model calibrated on the Istrana dataset, i.e. on field data only).

Numerical optimization was performed by the Simulated Annealing method (Kirkpatrick et al., 1983). Among the many existing versions of this algorithm, in the present study we used the one proposed by Corana et al. (1987), which in our implementation was driven by the Marsaglia pseudo-random number generator (Marsaglia et al., 1990), and programmed to minimize the Root Mean Squared Error (RMSE) between estimated and observed phenological event dates:

$$\mathrm{RMSE} = \sqrt{\frac{\sum_{i=1}^{n}(E_i - O_i)^2}{n}} \quad (2)$$

where E_i = ith estimated budburst, flowering or veraison date, O_i = the ith corresponding observed date and n = number of observations for each phenophase. The RMSE to be minimized was the average of the RMSEs calculated for budburst, flowering and veraison.

In CF models, minimum, optimal and maximum chilling-effective temperatures were searched in the following intervals: [−5; 5 °C], [0; 10 °C] and [0; 20 °C]. In both CF and F_{beta} models, optimal forcing temperatures were searched between 10 and 45 °C. Critical sums of chilling as well as forcing units were explored between 0 and 200 units. In $F_{beta/GDD}$ models, t_0 was searched between day-of-year (DOY) −60 and 90, where the negative sign indicates number of days before January 1st. In F_{GDD} models, T_b

was searched between 0 and 15 °C, and critical GDD sums between 0 and 1000.

Optimization was performed for each model separately, running the algorithm several times, each time changing the initial parameters.

2.4. Scenario analysis

Artificially-generated daily temperature series were obtained by downscaling HadCM3 predictions available as monthly means (Pope et al., 2007) for the Susegana site, by means of the LARS-WG5 software, which is a stochastic weather generator developed for simulating weather data at a single site under both current and future conditions (Racsko et al., 1991; Semenov and Stratonovitch, 2010).

The software was calibrated on local weather data from 1960 to 2010, to characterize the probability distributions of weather variables as well as correlations between them, and used to generate synthetic weather time series of the desired length by randomly selecting values from the appropriate distributions. Daily temperature series were obtained for the following periods: 2011–2033, 2046–2065 and 2080–2099. The period 1980–1999 was considered as reference baseline. The generated series, each 30 years long, were not true time series, but rather constituted replicates of a given year's weather, reproducing the variability of the historical dataset in the projected climate.

For the sake of simplicity, each period is referred to by its central year, i.e. 2020, 2055, 2090 and 1990.

We considered the IPCC emission scenario A2 (IPCC, 2007), which is one of the more pessimistic, predicting a global mean temperature increase of 3.4 °C by 2100.

All models were run on the climate projections to evaluate the effect of parameterization differences on the model response.

2.5. Statistical analysis

The adherence of estimates to external data was quantified with RMSE, the Nash-Sutcliffe Modeling Efficiency index (EF) (Nash and Sutcliffe, 1970), and Akaike's Information Criterion (AIC) (Burnham and Anderson, 2002).

RMSE (Eq. 2) quantifies the average deviation of estimates from observations in the same units (in this case, days).

EF (Eq. 3) represents the proportion of variation explained by the model. If its value is below zero it indicates that the average of observations is a better predictor than the model.

$$\mathrm{EF} = 1 - \frac{\sum_{i=1}^{n}(E_i - O_i)^2}{\sum_{i=1}^{n}\left(O_i - \overline{O}\right)^2} \tag{3}$$

The AIC was calculated with the following equation:

$$\mathrm{AIC} = n \cdot \ln\left(\frac{\sum (E_i - O_i)^2}{n}\right) + 2k \tag{4}$$

Unlike RMSE and EF, which give an immediate grasp of model performance as their units and variation interval have an intuitive meaning, AIC values and variation interval depend on the number of measurements and parameters, so it is very difficult to compare them with results of other studies, and to set standard threshold limits beyond which to judge the quality of response.

For this reason we calculated ΔAIC:

$$\Delta\mathrm{AIC} = \mathrm{AIC}_i - \mathrm{minAIC} \tag{5}$$

where AIC_i is the AIC calculated for the *i*th model, while minAIC is the "best" model, i.e. the one yielding the lowest value.

All the statistics were calculated for each individual phenological event, which were then averaged.

To facilitate model comparison, the three indices were aggregated in a single adimensional indicator, based on the principle of fuzzy logic, as described by Bellocchi et al. (2002). The indicator, termed FI (Fuzzy Indicator) varies between 0 (=good performance) and 1 (=bad performance).

The method has been applied in many other studies, such as Rivington et al. (2005), Donatelli et al. (2005), Diodato and Bellocchi (2007), Confalonieri et al. (2009) and Abraha and Savage (2008).

Only an essential and intuitive description will be given here as the interested reader can find all details in Bellocchi et al. (2002).

Each statistic was assigned two arbitrary thresholds, based on the authors' expertize, to classify the model response as "good", "bad" or "intermediate". RMSE values were considered "good" if they were less than 1 day, while they were "bad" if more than 10. Similarly, EF was considered "good" if higher than 0.8, and "bad" if less than 0. For ΔAIC we adopted existing practical guidelines, which consider "good" values < 2, and "bad" values above > 10 (Burnham and Anderson, 2002).

If all indices were in the "good" interval, FI was assigned a 0 score, while if they all fell in the "bad" interval, FI scored 1. The indicator took intermediate scores when one or more of the basic statistics values fell between the assigned thresholds. In this case, final values were calculated by means of an "S" shaped function, using aggregating rules based on the multi-valued fuzzy set theory introduced by Zadeh (1965), and the so-called Sugeno or Takagi–Sugeno–Kang method of fuzzy inference (Sugeno, 1985).

For testing model accuracy we utilized validation results, i.e. performance indices calculated on model outputs from independent dataset. We did not consider calibration results, as they were derived from differently sized datasets. Fitting results from the largest dataset are likely to be worse just because models have to fit more data, so they are not comparable to the other ones.

The effect of model and calibration on estimation accuracy was tested with a 2-way ANOVA, taking as dependent variable the difference between observed and estimated DOY for each phenophase. Normality of data was previously assessed with the Shapiro–Wilks test while homogeneity of variances was checked with the Levene test. All tests were performed with the R software v. 2.15.0 (R Development Core Team, 2012).

The ANOVA was also performed on scenario simulations. In this case, lacking observed data, the dependent variable was the estimated DOY for each phenological event. The analysis therefore did not assess accuracy differences but simply the differences among the predicted dates.

Individual effects were tested by orthogonal comparisons after ANOVA with the following scheme:

- CF vs. F models;
- F_{beta} vs. F_{GDD};
- One forcing T_{opt}/T_b vs. (2 and 3) T_{opt}/T_b;
- Two forcing T_{opt}/T_b vs. three T_{opt}/T_b.

3. Results

3.1. Growth-room experiments

The number of days elapsed from the removal of vine cuttings to budburst is displayed in Fig. 1 (left panel). Explants sampled until November 11th of the second year did not sprout, so they were assigned the value 0. In addition, the earliest samples collected in the first year took more than two months to reach 50% budburst,

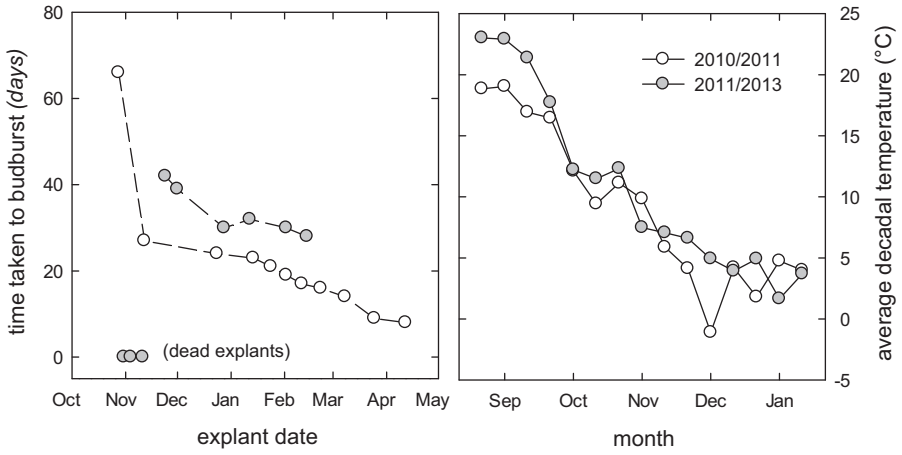


Fig. 1. Left panel: number of days elapsed between cutting removal from the field and budburst observation under controlled conditions. Right panel: decadal mean temperature recorded on the cuttings source field between September and January in 2010/2011 and 2011/2012. White and grey symbols refer to 2010/2011 and 2011/2012 respectively.

which was irregularly spread over a long time, with weak and thin shoots. Failed and excessively slow-sprouting samples were not utilized to calibrate models.

From mid-November onwards budburst was regular and increasingly faster. The explants taken in 2010/2011 showed an earlier budburst with respect to the following year, as the mean temperatures between September and December were almost always lower than in the same period in the following year (Fig. 1, right panel).

GR data allowed the range of climatic conditions to be extended with regard to the chilling period duration. In the September–February period, the number of days with $T_{mean} < 10\,^{\circ}C$ varied between 44 and 158 in GR data, while for Istrana and Susegana datasets the intervals were 147–161 and 147–168. GR also provided shorter winters than those expected in the future as in the A2 scenario projections the fewest number of chilling days was 130 (Table 3).

3.2. Model fitting and validation

Fig. 2 reports the results of the preliminary evaluation designed to assess the best T_{max} settings of the beta-type function. The displayed data are the average FI values obtained at different T_{max} settings of the beta-type function. The T_{max} value giving the best results and hereafter adopted was 35 °C for both calibrations, although the effect was more evident on the Istrana* dataset.

Using the Istrana dataset for calibration, fitting values of the FI varied between 0.008 and 0.356. Higher FI values were obtained with Istrana*, because the model had to fit more data. The variation interval here was between 0.036 and 0.442 (Table 4).

In both calibrations the models giving the best FI scores at validation belonged to the F_{beta} type, with the best performances obtained with the lowest complexity level, i.e., 0.368 and 0.263 for $F_{beta}5$ and $F_{beta}5$* respectively, against the maximum observed of 0.555 and 0.530 in Istrana and Istrana*. Models calibrated on Istrana* performed better than those on Istrana (Table 4).

However, as resulted from ANOVA (Table 5), only the calibration dataset had a significant effect on performance. Hence, we did not test individual effects.

Flowering was the event most accurately predicted by all models in both calibrations, with validation FI scores varying from 0.015 to 0.453 (Table 6). On the contrary, the worst results were for budburst, with scores varying from 0.248 to 0.819. Veraison showed an intermediate accuracy, between 0.204 and 0.614. On average, for all phenophases, the accuracy was higher in the models calibrated on the Istrana* dataset (Table 6).

Table 3
Variability of datasets. Days with $T_{mean} < 10^{\circ}$ in the September–February period.

	Min	Median	Max
Istrana	147	153	161
Susegana	147	159	168
Growth-Room	44	114	158
A2_2020	145	155	164
A2_2055	139	145	151
A2_2090	130	136	144

3.3. Consistency of parameterization

3.3.1. Chilling related parameters

CF models calibrated on the Istrana* dataset converged to more physiologically plausible chilling requirements than in the alternative calibration (Table 7). This is expressed by the C_{crit} parameter, which quantifies the number of days of exposure to the optimal chilling temperature needed to break endodormancy. In models calibrated on Istrana*, C_{crit} values ranged from 11.6 to 39.7 days,

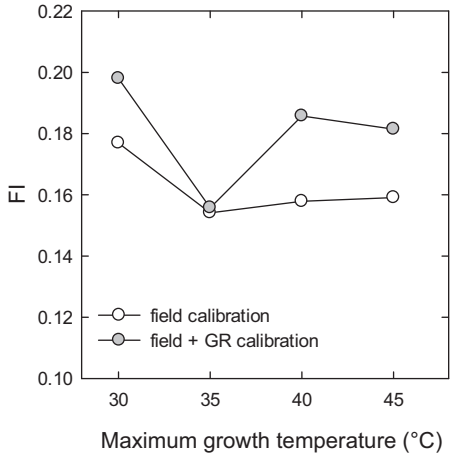


Fig. 2. Assessing the T_{max} parameter of the beta-function for calculating forcing units. The validation accuracy obtained at different T_{max} values was calculated as FI score, and averaged over all beta function-based models for each calibration dataset. The T_{max} yielding the minimum score was adopted.

Table 4
Fitting and validation results of the phenology models. Models marked with an asterisk (*) were calibrated on an extended phenology dataset, including budburst observations from growth-room experiments.

model	Fitting				Validation			
	RMSE	EF	ΔAIC	FI	RMSE	EF	ΔAIC	FI
CF8	2.65	0.88	5.8	0.008	5.28	0.56	19.1	0.540
CF9	2.65	0.88	8.0	0.287	5.12	0.59	11.3	0.520
CF10	2.63	0.88	9.6	0.322	5.23	0.58	15.4	0.541
$F_{beta}5$	2.70	0.87	0.6	0.009	5.24	0.57	5.6	0.368
$F_{beta}6$	2.65	0.88	2.0	0.014	5.20	0.57	6.9	0.383
$F_{beta}7$	2.66	0.88	4.2	0.092	5.19	0.58	8.7	0.511
$F_{GDD}5$	2.94	0.84	4.2	0.157	5.50	0.53	10.8	0.528
$F_{GDD}6$	2.94	0.84	5.7	0.356	5.44	0.54	10.8	0.523
$F_{GDD}7$	2.82	0.86	6.5	0.223	5.35	0.55	10.9	0.555
CF8*	4.77	0.82	10.3	0.247	5.37	0.55	19.1	0.530
CF9*	4.00	0.86	3.0	0.074	4.92	0.62	8.4	0.459
CF10*	4.38	0.83	9.0	0.036	5.36	0.53	17.0	0.519
$F_{beta}5$*	4.68	0.84	11.0	0.281	5.09	0.60	3.1	0.263
$F_{beta}6$*	3.93	0.85	1.9	0.047	5.00	0.61	3.9	0.289
$F_{beta}7$*	3.92	0.86	1.8	0.043	4.82	0.63	5.4	0.304
$F_{GDD}5$*	4.86	0.77	13.4	0.442	5.28	0.59	7.3	0.314
$F_{GDD}6$*	4.42	0.81	8.0	0.324	5.25	0.57	7.7	0.425
$F_{GDD}7$*	4.48	0.81	7.6	0.345	5.20	0.58	9.0	0.482

Table 5
ANOVA of data from Table 4.

Source of variation	Df	Sum Sq	Mean Sq	*F* value	Pr(>*F*)
Model	8	94	11.71	0.4798	0.8713
Calibration	1	2654	2653.92	108.7443	<0.001***
Model × calibration	8	98	12.27	0.5029	0.8549
Residuals	2196	53594	24.41		

Table 6
Validation FI scores for each phenological event estimated by models under study. Models were calibrated on field data ("Istrana" dataset, left side) and on the same data integrated by budburst observation from growth room (GR) experiments ("Istrana*" dataset, right side).

Istrana calibration				Istrana* calibration			
Model	Budburst	Flowering	Veraison	Model	Budburst	Flowering	Veraison
CF8	0.700	0.367	0.553	CF8*	0.705	0.393	0.494
CF9	0.657	0.324	0.580	CF9*	0.609	0.350	0.419
CF10	0.659	0.373	0.592	CF10*	0.819	0.376	0.362
$F_{beta}5$	0.630	0.192	0.283	$F_{beta}5$*	0.261	0.129	0.400
$F_{beta}6$	0.685	0.088	0.378	$F_{beta}6$*	0.484	0.177	0.206
$F_{beta}7$	0.644	0.306	0.585	$F_{beta}7$*	0.693	0.015	0.204
$F_{GDD}5$	0.694	0.420	0.470	$F_{GDD}5$*	0.248	0.453	0.241
$F_{GDD}6$	0.703	0.393	0.474	$F_{GDD}6$*	0.687	0.195	0.392
$F_{GDD}7$	0.682	0.370	0.614	$F_{GDD}7$*	0.661	0.379	0.407

Table 7
Chilling-related parameters obtained after fitting of CF models. T_{min}, T_{opt} and T_{max} are the cardinal temperatures (°C) in the chilling-effective interval. C_{crit} is the chilling requirement, quantifying the number of days at the optimal chilling temperature that must be experienced by plants to overcome endodormancy.

Model	T_{min}	T_{opt}	T_{max}	C_{crit}
CF8	−4.6	6.2	19.7	122.4
CF9	−4.2	5.9	19.9	117.6
CF10	−3.7	4.7	19.7	115.3
CF8*	−4.6	2.2	19.0	39.7
CF9*	−0.6	0.4	19.1	30.3
CF10*	−1.7	1.7	13.2	11.6

while in the alternative case the variation was between 115.3 and 122.4.

The two calibrations also yielded different chilling-effective temperatures. Models calibrated on the Istrana dataset had optimal temperatures between 4.7 and 6.2, while in the Istrana* calibration they varied between 0.4 and 2.2. The min–max intervals almost matched the explored intervals, from around −5 to 20 °C. Only in CF10* the interval was appreciably narrower, from about −1.7 to 13.3 °C.

3.3.2. Forcing-effective temperatures

In CF models calibrated on Istrana, optimal temperatures converged to 26–26.6 °C whatever the number of T_{opt} adopted. Similar behavior was observed in F_{beta} models, where the interval of variation was a little wider, between 25.0 and 26.8 °C (Table 8).

Differences related to the phenophase-specificity emerged on beta-function-based models when they were calibrated on the Istrana* dataset. In CF9* and CF10*, T_{bb} was 23.3 and 22.1 °C respectively, i.e., lower than $T_{fl\text{-}ve}$ (26.7 °C) or T_{fl} and T_{ve} (27.7° and 26.3°),

Table 8
Forcing-related parameters obtained for all models after optimization on Istrana or Istrana* datasets. T_{opt} is the optimal temperature in models using the beta-type function for calculating forcing units, while T_b is the base temperature to calculate GDD units in F_{GDD} models. The "t_0" column reports the starting date of forcing accumulation for F_{beta} and F_{GDD} models. For comparison, the date when end-of-endodormancy (EOE) occurs in CF models is also reported.

Model	T_{opt} or T_b					EOE	t_0	Critical forcing sums		
	bb-ve	bb	fl-ve	fl	ve			S_{bb}	S_{fl}	S_{ve}
CF8	26.4					46.3		5.8	23.5	50.4
CF9		26.1	26.6			40.5		6.8	22.2	49.3
CF10		26.0		26.3	26.0	39.3		7.1	23.4	51.8
CF8*	25.2					−34.3		13.1	28.9	55.7
CF9*		23.3	26.7			−34.8		24.1	21.8	48.8
CF10*		22.1		27.7	26.3	−45.3		35.1	17.0	50.4
$F_{beta}5$	26.2						54.2	6.2	24.0	51.3
$F_{beta}6$		25.6	26.4				53.1	7.7	23.0	50.1
$F_{beta}7$		25.0		26.8	25.9		48.3	9.6	20.6	52.1
$F_{beta}5$*	25.4						−3.8	10.9	27.8	52.8
$F_{beta}6$*		22.7	26.4				−17.8	24.0	22.7	50.5
$F_{beta}7$*		22.9		27.0	26.4		−14.2	24.7	16.7	58.5
$F_{GDD}5$	5.9						56.7	138.2	481.3	974.0
$F_{GDD}6$		5.5	8.4				56.2	149.8	352.3	825.5
$F_{GDD}7$		5.5		7.8	1.4		55.2	153.0	389.9	1239.5
$F_{GDD}5$*	3.8						−7.7	297.5	597.9	1092.6
$F_{GDD}6$*		2.9	7.4				−17.4	389.7	406.8	880.5
$F_{GDD}7$*		2.6		7.3	3.6		−16.0	409.4	414.9	1109.4

while in CF8* $T_{bb\text{-}ve}$ converged to 25.2 °C. In $F_{beta}6$* and $F_{beta}7$* T_{bb} was similarly 22.7 and 22.9 °C, with $T_{fl\text{-}ve}$ of 26.4 and T_{fl}–T_{ve} equal to 27.0 and 26.4 °C. In $F_{beta}5$ and $F_{beta}5$*, the unique $T_{bb\text{-}ve}$ resulted as being 26.2 and 25.4 °C respectively (Table 8).

Although T_b in F_{GDD} models does not have the same physiological meaning as T_{opt} in beta-based models, an analogous differentiation appeared, with a T_{bb} value lower than T_{fl} and $T_{fl\text{-}ve}$. Contrary to what was observed in the beta-based models, temperature variation was obtained in both calibrations, and T_{ve} was lower than T_{fl}.

3.3.3. Starting date of forcing accumulation

The calibration dataset dramatically influenced the estimated date of endodormancy break in CF models (Table 8, "EOE" column). In CF* models this event fell within the second half of November, between 34.3 and 45.3 days before January 1st, whereas in those optimized on Istrana it fell much later, around mid-February (39.3–46.3 DOY).

In forcing models the t_0 was likewise affected by the calibration dataset. With Istrana, t_0 varied between 48.3 and 56.7 DOY, while with Istrana* the variation interval was 17.8 to 3.8 days before January 1st.

3.4. Scenario analysis

The expected temperature rise with warming, as predicted by HadCM3, increased from autumn to summer (Table 9).

All models predicted a progressive shift to earlier onset of all the phenophases considered (Fig. 3). By 2090, budburst was predicted to occur from 13.9 (CF8) to 44.4 ($F_{GDD}6$*) days earlier than in 1990. For flowering, the time shift spanned between 24 (CF8) and 36.5 ($F_{GDD}5$*) days, and for veraison between 29 (CF10) and 43 days ($F_{GDD}6$*) (not shown).

Table 9
Variations of the mean seasonal temperatures for the SRES-A2 emission scenario and three temporal projections based on the HadCM3 model, downscaled to the Susegana site. Variations are relative to the baseline period 1980–1999.

Projection	Autumn	Winter	Spring	Summer
2020	0.0	+0.8	+0.9	+1.0
2055	+1.5	+1.7	+2.2	+3.1
2090	+3.2	+3.7	+4.4	+6.2

The calibration type was the more important source of variation in model response. Models calibrated on the Istrana dataset always predicted a smaller advance in a phenological stage than their counterparts.

The difference between the two types of calibration was attenuated on flowering and veraison (Fig. 3). For the 2090 projection, the difference between the averages calculated across all models for each calibration was 6.6 and 5 days in flowering and veraison respectively, while for budburst it was 17.3 days.

The calibration type had a significant effect in all projections, while models differentiated significantly only from 2055 onwards. By the 2090 projection the interaction model × calibration became significant (Table 10).

The post-hoc planned comparisons were performed for the 2090 projection (Table 11). CF models differed from $F_{beta/GDD}$ models for all phenophases and in both calibrations, predicting a higher shift from around 2 days for flowering to around 5 days for budburst.

Another significant difference was found between F_{beta} and F_{GDD} models, but the effect was different in the two calibrations, except for veraison.

All the other comparisons, designed to test the effect of complexity level, were mostly not significant. Where present, significance was detected on some phenophases only, with an evident interaction effect with calibration.

The differential variation predicted for phenological events, especially between budburst and flowering, turned into a variation in the length of the budburst–flowering and flowering–veraison intervals. The budburst–flowering interval at the 2090 projection resulted longer than in 1990 when estimated with models calibrated on the Istrana* dataset (Table 12a). The increment was between +6.5 and +8.6 days with CF* models and $F_{beta}5$* and $F_{GDD}5$*. The other models predicted higher increments, between +11.0 and +11.8 days. The lengthening was caused by the earlier prediction of budburst with respect to flowering. As a result, the average temperature experienced during this period decreased from −0.3 to −1.2 °C, as the sprouting process shifted towards the colder part of the season (Table 12b).

In the Istrana* calibration, F_{beta}*and F_{GDD}* models with one T_{opt}/T_b the increment of the budburst–flowering interval is shorter than in the models with higher phenophase specificity (Table 12). The latter ones have an earlier t_0 and a lower T_{opt}/T_b, which allow them to start forcing accumulation earlier than the former models.

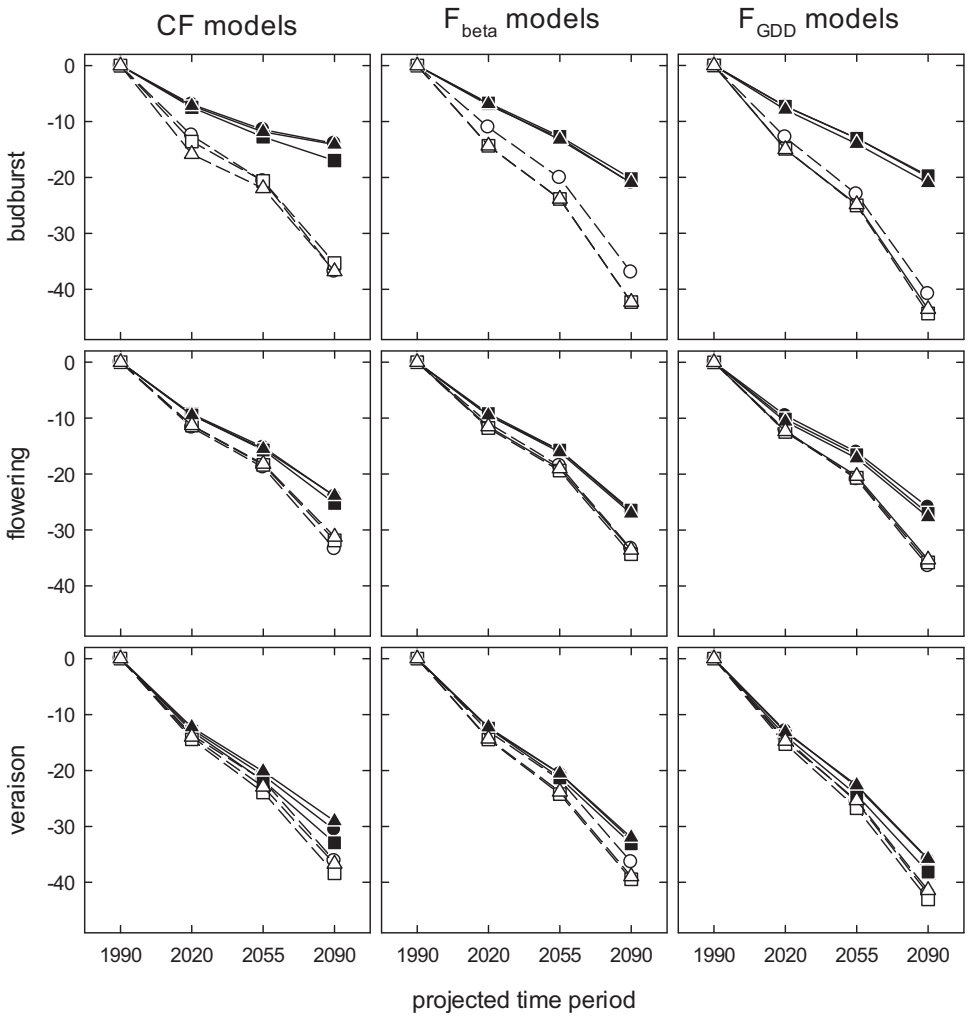


Fig. 3. Relative variation (days) of the timing of budburst, flowering and veraison predicted by CF, F_{beta} and F_{GDD} models for the A2 scenario, 1990–2090 period. Circles, squares and triangles stand for models with one, two or three optimal forcing temperatures respectively. Black and white symbols indicate calibration on the Istrana or Istrana*dataset respectively.

Under present climate conditions, both type of models, e.g. $F_{beta}5^*$ vs. $F_{beta}6^*/F_{beta}7^*$, start forcing accumulation in late Winter, to end up more or less at the same time (not shown). At the 2090 projection, $F_{beta}5^*$ starts forcing after 1 January, while at this time the other two models have accumulated more than the 20% of the total forcing heat required to sprout (not shown). This leads to an earlier sprouting which turns out into a longer budburst–flowering interval.

A quite different behavior was observed in models calibrated only on Istrana, where the budburst–flowering period shortened. This was more pronounced in CF models, at between 5 and 6.7 days, while in the others it was between 1.4 and 3.4 days. Thanks to a later budburst, which delayed the sprouting time, the budburst–flowering period was shifted later in the season, so the average temperature relative to 1990 increased between 0.7 and 1.8 °C (Table 12a and b).

The flowering–veraison interval resulted shorter in both calibrations, except for CF8* and $F_{beta}5^*$, with the highest variation observed in models calibrated on Istrana. The differences among models were lower than for the previous interval: on average the observed decreases were 1.4 and 3 days for Istrana* and Istrana-calibrated models respectively (Table 12a). Here the average temperature variation with respect to 1990 was higher than for the previous interval, but the predicted increase was about 1 °C lower in models calibrated on Istrana*, with very little difference among models in both calibrations (Table 12b).

During the budburst–flowering interval, the occurrence of days with supra-optimal temperatures, in the present time or in the future was detected only by models calibrated on Istrana, at 0.1% in the 2090 projection (not shown).

Higher occurrences were found for the flowering–veraison interval in both calibrations. Fig. 4 illustrates the results for the 2090

Table 10
P-values from two-way ANOVA testing the effect of model and calibration on the prediction of phenological events under future climate change scenarios.

Sources of variation	Budburst			Flowering			Veraison		
	2020	2055	2090	2020	2055	2090	2020	2055	2090
Model	0.249	0.003**	<0.001***	0.127	<0.001***	<0.001***	0.249	<0.001***	<0.001***
Calibration	<0.001***	<0.001***	<0.001***	0.003**	<0.001***	<0.001***	<0.001***	<0.001***	<0.001***
Model × calibration	0.087	0.063	<0.001***	0.243	0.171	<0.001***	0.087	0.568	0.177

Table 11
Two-way ANOVA post-hoc tests (orthogonal contrasts) to assess the effect of model type, complexity level and beta-function vs. GDD-based forcing units. The test has been performed on the 2090 projection estimates. The table reports the average difference in days between the two treatments being compared and the associated *P*-value.

Contrast	Calibration	Budburst		Flowering		Veraison	
		ΔDays	*P*-value	ΔDays	*P*-value	ΔDays	*P*-value
CF vs. (F_{beta} + F_{GDD})	Istrana	4.94	<0.001***	2.44	<0.001***	3.44	<0.001***
	Istrana*	5.54	<0.001***	2.80	<0.001***	3.01	<0.001***
F_{beta} vs. F_{GDD}	Istrana	−1.37	<0.001***	−0.09	<0.001***	3.26	<0.001***
	Istrana*	1.80	<0.001***	2.04	<0.001***	3.02	<0.001***
CF8 vs. (CF9 + CF10)	Istrana	0.64	0.999	0.58	0.998	0.56	0.992
	Istrana*	−0.78	1.000	−2.92	<0.001***	0.42	0.999
$F_{beta}5$ vs. ($F_{beta}6$ + $F_{beta}7$)	Istrana	−0.03	1.000	−0.03	1.000	0.11	1.000
	Istrana*	4.78	0.997	0.50	0.997	1.58	0.761
$F_{GDD}5$ vs. ($F_{GDD}6$ + $F_{GDD}7$)	Istrana	0.25	1.000	0.75	0.917	0.42	0.999
	Istrana*	2.64	0.002**	−1.36	0.173	0.19	<0.001***
CF9 vs. CF10	Istrana	−2.83	0.051.	−1.06	0.740	−3.22	<0.001***
	Istrana*	−0.11	1.000	−1.17	0.601	−1.72	0.085
$F_{beta}6$ vs. $F_{beta}7$	Istrana	0.83	1.000	0.72	1.000	−0.78	1.000
	Istrana*	0	0.782	−1.56	1.000	−0.39	0.009**
$F_{GDD}6$ vs. $F_{GDD}7$	Istrana	0.17	1.000	−0.50	0.999	−2.28	0.005**
	Istrana*	−0.39	<0.001***	−0.28	1.000	−0.94	<0.001***

Table 12
Variation in length (a) and daily mean temperature (b) of phenological intervals in the 2080–2099 climate projection with respect to the baseline period 1980–1999. (BB-FLO = budburst–flowering; FLO-VE = flowering–veraison).

Model	BB-FLO	FLO-VE	Model	BB-FLO	FLO-VE
(a) Variation of interval length (Δ days)					
CF8	−6.7	−2.8	CF8*	6.5	1.1
CF9	−5.0	−3.8	CF9*	6.6	−2.6
CF10	−6.5	−1.3	CF10*	8.6	−1.6
$F_{beta}5$	−2.4	−1.8	$F_{beta}5$*	6.7	0.8
$F_{beta}6$	−3.0	−2.8	$F_{beta}6$*	11.0	−1.3
$F_{beta}7$	−2.8	−1.1	$F_{beta}7$*	11.8	−1.5
$F_{GDD}5$	−2.7	−6.1	$F_{GDD}5$*	7.7	−1.9
$F_{GDD}6$	−1.4	−3.4	$F_{GDD}6$*	11.7	−3.4
$F_{GDD}7$	−3.4	−4.2	$F_{GDD}7$*	11.4	−2.3
(b) variation of daily mean temperature (Δ°C)					
CF8	+1.8	+3.6	CF8*	−0.6	+2.5
CF9	+1.5	+3.3	CF9*	−0.3	+2.6
CF10	+1.8	+3.7	CF10*	−0.4	+2.7
$F_{beta}5$	+1.1	+3.2	$F_{beta}5$*	−0.5	+2.5
$F_{beta}6$	+1.2	+3.2	$F_{beta}6$*	−1.0	+2.3
$F_{beta}7$	+1.0	+3.2	$F_{beta}7$*	−0.9	+2.4
$F_{GDD}5$	+1.2	+3.1	$F_{GDD}5$*	−1.1	+1.9
$F_{GDD}6$	+0.7	+2.7	$F_{GDD}6$*	−1.2	+2.0
$F_{GDD}7$	+1.1	+3.0	$F_{GDD}7$*	−1.1	+2.1

projection, where the percentage was between 21.8% and 37.8% relative to the 2.3–10.0% range of present conditions.

With the only exception of CF8 and $F_{beta}5$, models calibrated on Istrana predicted higher hot-days percentages than those calibrated on Istrana*, with a lower differentiation among them.

These differences between the two types of calibration are entirely attributable to the variation in the onset of forcing accumulation, which, in CF models coincides with the end of the endodormancy phase.

In CF models calibrated on Istrana, the predicted endodormancy break date is DOY 43 on average. By this time of the year, favorable thermal conditions have already taken place by some weeks. Therefore, a later forcing accumulation beginning cause models to neglect days where phenological development could potentially take place, thereby underestimating budburst timing. Conversely, models which predict an earlier exit of endodormancy have a lower probability to underestimate budburst timing, as they are "ready" to capture favorable thermal conditions long before spring temperature rise.

In the forcing models the calibration effect on t_0 was analogous to that observed on endodormancy exit, since field calibration made models to converge towards a late forcing accumulation starting, with the same consequences as in CF models. Likewise, models with an earlier t_0 showed the capacity of capturing favorable thermal conditions as soon as they establish.

This last concept is visually exemplified in Fig. 6. The two arrows indicate the mean starting point of forcing units accumulation, averaging across CF and F models which was −21 for Istrana* calibration and DOY 50 for the other case.

Starting dates predicted by Istrana* calibration preceded spring thermal increase in present conditions as well as in the future. In the alternative case, starting date preceded spring thermal increase in present conditions but not in the future. The fraction of cumulated heat missed by the latter calibration caused underestimation of budburst timing.

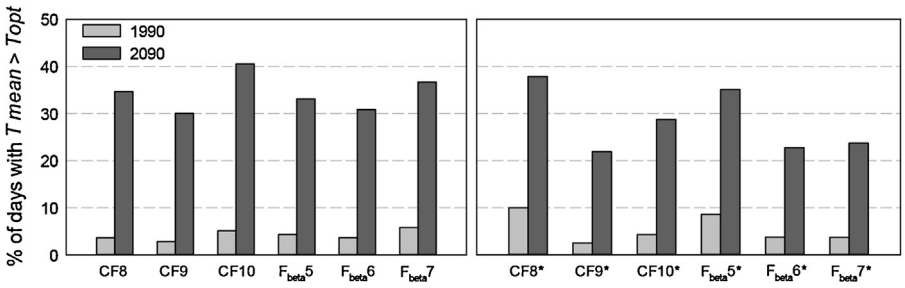


Fig. 4. Proportion of days with mean temperature above T_{opt} predicted by models using the beta-type function for the flowering–veraison interval, under present conditions and the 2090 climate projection.

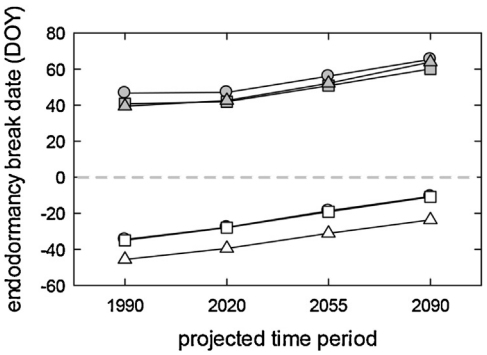


Fig. 5. Variation of the end-of-dormancy date, as predicted for present conditions and future projections (A2 scenario) by CF models calibrated on the Istrana or the Istrana* dataset. Circles, squares and triangles stand respectively for models with one, two or three optimal or base temperatures for the forcing phase. Grey and white symbols indicate calibration on Istrana or Istrana* dataset respectively.

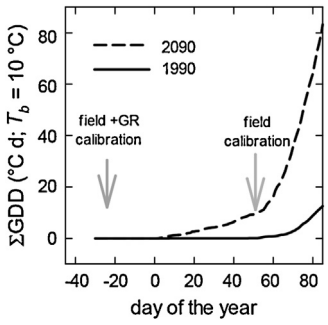


Fig. 6. Average cumulated GDD curves (T_b = 10 °C) calculated for present conditions (Istrana dataset, 1980–1999) and for the A2 scenario prediction for the same site and for the 2080–2099 period. Grey arrows indicate the average starting point of forcing accumulation estimated by Istrana and Istrana* calibrated models.

4. Discussion

4.1. Growth-room experiments

Although cuttings-based GR observations have long been used for phenological studies (e.g. Antcliff and May, 1961; Pouget, 1964; Or et al., 1999, to mention only studies on grapevine), they represent a rather different system from whole plants in the field, with predictably different responses to environmental variables. For this reason their use in parameterizing models to be applied to field-application could be questionable. This was investigated in a previous study (Fila et al., 2012), where it was demonstrated that models calibrated only on GR data provided a good accuracy when tested on two different field-based datasets. The present results corroborate these previous findings, showing that integrating GR data into the calibration dataset yielded more accurate estimates than those obtained with field-calibrated models (Table 4).

Both field-based phenological datasets used in this study were characterized by long and cold winters, where budburst was never observed before mid-March. In order to properly calibrate chilling parameters, a higher variability was desirable to raise the probability that the optimization process was constrained around the critical temperature intervals. To this end, we obtained additional data by conducting budburst observations on variously chilled vine cuttings forced to sprout in a growth room.

Using budburst observations derived from artificially forced vine cuttings increased constraints in the calibration dataset and oriented optimization towards more consistent parameters (Table 7).

Lack of sprouting in some of the earliest samples was most likely due to insufficient fulfillment of chilling requirements at the sampling date, and this indicates that the time interval explored by the extended dataset was wide enough to encompass all the dormancy status.

4.2. Model fitting and validation

The proposed CF models, despite their wider explanatory capability were not able to outperform the simpler forcing models in terms of external accuracy, i.e. the capacity to provide accurate estimates on data independent of those they were calibrated against (Table 4).

As explained in the introduction, this could well depend on overfitting effects caused by the higher parameter numbers, and on the fact that calibration was performed on a dataset with only long winters, where the exact determination of dormancy end was not critical to start capturing forcing-effective days. However, it cannot be excluded that the modelling approaches used here were not able to capture unknown effects taking place when the onset of thermal accumulation falls early in the season. This is discussed below in Section 4.4.

Budburst and veraison proved to be the events most difficult to predict (Table 6). The higher discrepancies between estimates and observations may have had multiple causes. Despite the standardized criteria used to visually determine the exact onset date, some lack of homogeneity between the two historical datasets is always possible. For budburst, an effect of winter pruning adding variability to the event also cannot be excluded (Friend and Trought, 2007; Martin and Dunn, 2000; Valenti et al., 2011). For veraison, a lower estimation accuracy may depend in part on the inaccuracy of visual assessment, which is trickier for white grape. Another possibility is that the effect of environmental variables such as water stress (Castellarin et al., 2007) or extreme temperatures (Mori et al., 2007; Sadras and Moran, 2012; Carbonell-Bejerano et al., 2013), which are both more likely to occur in this period, may have impaired the temperature dependency of pigments accumulation in berry skin, thus adding uncertainty to model predictions.

From a purely statistical viewpoint, all models showed comparable predictive abilities, so each of them could safely be used for applications targeting current climate conditions. However, adopting metrics evaluating “quality” criteria of modelling, as the AIC, simpler models have reached higher scores, meaning that under equal predictive ability, the most simple models are to be preferred (Table 4). Among these, those adopting 5 parameters, the ones with curvilinear description of temperature dependency of forcing rate are the best choice.

4.3. Consistency of parameterization

Beyond their utility for making predictions for practical as well as scientific purposes, models are fundamental tools to generalize experimental information and advance scientific knowledge. The value of a model depends therefore on the extent to which the quantitative information expressed by its parameters adheres to the real properties of a given system, thus contributing to increase our knowledge about it.

Moreover, in the absence of observations against which to assess model accuracy, parameterization consistency offers the only clue to deduce how reliable a given model is outside the boundary conditions within which it has been calibrated. In that respect, our results showed that while all models under study had comparable accuracy, under present day conditions, they showed remarkable differences in their parameterization, which was mostly affected by the type of calibration (Tables 7 and 8).

Although our interest was related to scenario analysis, the same considerations apply for current climate conditions in different zones from where the models have been calibrated. Under current climate conditions, both parameterizations worked well, but this is valid for the local conditions. If the models are to be applied elsewhere, the parameterizations differences should be therefore taken into account, and the one with the highest chances of capturing different situations is the one to be recommended.

4.3.1. Chilling-related parameterization

The most striking variation among differently calibrated models concerns chilling requirement, quantified by the C_{crit} parameter in CF models (Table 7). According to the classical criterion for defining chilling-effective days, i.e. days with $T_{mean} < 10\,^{\circ}C$ (Dokoozlian, 1999), models calibrated on Istrana* had to fit chilling exposure as short as 44 days, while the shortest chilling duration in Istrana was 147 days (Table 3). As a result CF models calibrated on the latter dataset required up to four months of exposure to chilling temperatures. This is in contrast with the available experimental evidence (Lavee and May, 1997; Dokoozlian, 1999; Botelho et al., 2007) and even with our experimental results, which demonstrated that grapevine has a low chilling requirement. GR experiments clearly show the capacity of cuttings to sprout early when placed under adequate temperature conditions. The chilling requirement obtained by the Istrana* dataset appears to be more plausible.

The optimized chilling-effective temperature range poses some interpretation challenges, because, according to this result, temperatures above 10 °C appeared to be effective in chilling fulfillment, in contrast with available experimental knowledge.

What we hypothesize is that during optimization a trade-off took place between C_{crit} and the width of chilling-effective temperature range. With the Istrana dataset, where models converged to high C_{crit}, a wide interval was expected because this allows more days to be chilling-effective. Conversely, when models were forced against short winters, yielding low C_{crit} values, a narrowing of the chilling temperature interval was expected as well because this is associated with fewer chilling-effective days. Indeed this was the behavior observed with CF10*. In CF8* and CF9* we did not obtain a narrowing of the interval, but rather a shift of T_{opt} towards 0 °C, which has a similar effect though, as days with lower temperatures are assigned a higher chilling effect. This could be interpreted as a partial convergence towards the expected result.

It should be noted that the optimized CF10* was almost equivalent to CF9*, having similar forcing temperatures for budburst and for flowering–veraison (Table 8). The reason why at least CF9* did not show the same behavior is therefore to be attributed to the subtle difference in the dynamics of the numerical optimization, which could be obviated by increasing the number of observations.

4.3.2. Forcing-related parameterization

All CF and F_{beta} models converged to very similar parameter values when optimization was performed against Istrana dataset, irrespective of model complexity, while different optimal temperatures were found at increasing phenophase specificity when the models were optimized on Istrana* (Table 8). The highest complexity level therefore did not add any information about temperature dependency of development, so one temperature forcing requirement for budburst and one for the following developmental phases were sufficient to describe it.

A slightly more complex behavior was shown by F_{GDD} models, where the specific T_{ve} in $F_{GDD}7$ and $F_{GDD}7$* took a value lower than T_{fl}, unlike in $F_{beta}7$* model (Table 8). This is difficult to justify with available physiological knowledge as there is no reason to support a lower temperature requirement for veraison, so an overfitting effect is the most plausible explanation.

4.4. Starting day of forcing accumulation

The factor appearing to most affect the response of models under future conditions was the timing of forcing accumulation onset.

When only field data were used to calibrate forcing models, fitting was optimal with t_0 values falling in late winter or early spring, from 48.3 to 56.7 DOY (Table 8). Similar results have been obtained by Parker et al. (2011), with $t_0 = 60$, Williams et al. (1985) with $t_0 = 51$ DOY, and Duchêne et al. (2010) with $t_0 = 46$. Nendel (2010) also obtained an average t_0 of 60 with a Spring Warming model where forcing units were calculated via a triangle-shaped function. This latter author reports that March 15th had already been proposed as t_0 by Alleweldt and Hofäcker (1975; cited by Nendel).

As proposed by Linkosalo and Lechowicz (2006), the fixed date for the start of temperature sum accumulation could indicate the existence of some other factor characterized by constant periodicity, e.g. a photoperiod effect, which would be needed to trigger post-dormancy development. This hypothesis was not explored because a photoperiod effect on grapevine phenology was experimentally demonstrated only for dormancy induction (Schnabel and Wample, 1987; Fennel and Hoover, 1991). For CF models, this is consistent with the idea of assuming a fixed calendar date (September 1st) as the starting moment to accumulate chilling units. No photoperiod action has so far been demonstrated on post-dormancy physiology of grapevine (Nendel, 2010).

Furthermore, the photoperiod hypothesis is not consistent with our results, which showed that t_0 was largely affected by the variability of the calibration dataset, and that a t_0 near the traditional setting of January 1st performed better than later onset times.

Other potential causes should therefore be looked for to explain why optimization converges to late starting dates of forcing onset. Since solutions implying early starting dates were discarded when the algorithm was run on datasets with long winters, it can be deduced that an earlier thermal accumulation generated a higher error. In other words, thermal accumulations starting early, e.g. on occasional mild days during winter, were not effective. More complex processes are therefore likely to be involved in the transition from dormancy to budburst than those described by the models used here, which need to be clarified and implemented in more advanced models. Fishman et al. (1987) proposed a more sophisticated model for post-dormancy development, where thermal accumulation is a two-phase process, each with specific thermal requirements and critical thresholds, and where the accumulation of a thermal unit requires meeting requirements of both. This model is very sophisticated, is hourly based and difficult to apply to most available datasets. However, the concepts are valid, and could be investigated with specific experiments.

4.5. Scenario analysis

All CF and SW models predicted a shift towards earlier onset of all the phenological phases under future climate scenarios (Fig. 3). Predictions obtained with models calibrated only on field data diverged progressively from those obtained with short winters in the calibration dataset. A delay in endodormancy break postponed the onset of forcing accumulation, so a parallel delay of budburst should be expected (Lavee and May, 1997; Webb et al., 2007). By the 2090 projection, dormancy break delay was about 20 days in both Istrana and Istrana* calibrated CF models. In the latter however, the delay is entirely within December, i.e. before forcing accumulation takes place (Fig. 5), and therefore has no effect in delaying forcing accumulation onset.

The shift discrepancy between the calibration types was already observed for budburst in a previous study (Fila et al., 2012). A confirmatory response was therefore expected for flowering and

veraison, which was indeed observed, but to a much lesser extent than for budburst, even at the 2090 projection.

The reason why the difference was attenuated is rather intuitive: thermal accumulation is slower when it begins earlier, because temperatures are lower. Therefore, the "advantage" of models reaching budburst earlier tended to reduce as development proceeds, as later-starting models accumulated forcing units faster.

In the absence of direct phenological observations in conditions similar to those expected in the future, there is no way to assess *a priori* which model should be recommended for future estimates. A simple reasoning however suggests that models which do not properly take into account the actual dormancy duration, and which postpone forcing accumulation onset in late winter or spring are likely to underestimate phenological shift due to climate change.

In our analysis, limited to the A2 scenario, this effect already became evident with the 2020 projection, at least for budburst. For flowering and veraison the discrepancy became appreciable only at the remotest projections.

The variability of calibration dataset resulted as the factor which most affected model response in the future. Variation due to model structure could be observed starting from the 2055 projection, with a clear effect of dormancy description and the shape of temperature dependency of post-dormancy development rate, while the presence of interactions makes it difficult to interpret the effect of the complexity level.

The variation in the timing of key events impacted on the thermal regime experienced by plants, as the various descriptions offered by models were not equivalent from a physiological point of view. Models based on the beta-type function, which discriminate the effect of temperatures on development rate according to their distance from T_{opt}, allowed a more physiologically insightful characterization of development in relation to the thermal regime experienced (Fig. 4).

Some differentiation of model complexity emerged here, especially in the Istrana* calibration, with more complex models predicting a higher incidence of unfavorable conditions in projected scenarios.

5. Conclusions

With respect to our initial questions the following conclusions can be drawn from the results outlined above:

- Under present climatic conditions, all models under study gave comparable results in terms of estimation accuracy. In particular, modelling solutions based on the CF model paradigm were not superior to the simpler forcing models. For applications targeted at present conditions, the latter should be therefore the recommended option, with preference given to models using the beta function.
- When aspects other than accuracy were considered, the factors under study were found to induce noteworthy differences:
  - In CF models, calibration datasets including budburst observation from GR experiments allowed more consistent chilling-related parameterizations and endodormancy duration estimates than with field data alone, thanks to the wider range of conditions that offered better constraints to orient model optimization.
  - Fitting on field + GR data had a remarkable impact on the forcing accumulation onset in both CF and F models. This represented the main source of variation in model behavior under a future climate scenario, more than the model structure itself.
  - Under a climate change scenario it was possible to observe a differential behavior attributable to the model mathematical definition only at a later projection.
- Care should be taken when models optimized on present time data are used for climate change impact analysis. This models are highly likely to underestimate phenological shifts, as they estimate thermal accumulation onset to occur later than actual spring temperature rise.

The study evidenced the need for a better understanding of factors other than temperature potentially affecting phenology, notably photoperiod, as well as the need of model improvements to account for variation in the temperature effect after endodormancy is broken.

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References

Abraha, M.G., Savage, M.J., 2008. Comparison of estimates of daily solar radiation from air temperature range for application in crop simulations. Agric. Forest Meteorol. 148, 401–416.
Alleweldt, G., Hofäcker, W., 1975. Einfluß von Umweltfaktoren auf Austrieb, Blüte, Fruchtbarkeit und Triebwachstum bei der Rebe. Vitis 14, 103–115.
Amaducci, S., Colauzzi, M., Bellocchi, G., Venturi, G., 2008. Modelling post-emergent hemp phenology (*Cannabis sativa* L.): theory and evaluation. Eur. J. Agron. 28, 90–102.
Antcliff, A.J., May, P., 1961. Dormancy and bud burst in Sultana vines. Vitis 3, 1–14.
Bellocchi, G., Acutis, M., Fila, G., Donatelli, M., 2002. An indicator of solar radiation model performance based on a fuzzy expert system. Agron. J. 94, 1222–1233.
Bock, A., Sparks, T., Estrella, N., Menzel, A., 2011. Changes in the phenology and composition of wine from Franconia, Germany. Climate Res. 50, 69–81.
Bonhomme, R., 2000. Bases and limits to using 'degree.day' units. Eur. J. Agron. 13, 1–10.
Botelho, R.V., Pavanello, A.P., Pires, E.J.P., Terra, M.M., Muller, M.M.L., 2007. Effects of chilling and garlic extract on bud dormancy release in Cabernet sauvignon grapevine cuttings. Am. J. Enol. Vitic. 58, 402–404.
Burnham, K.P., Anderson, D.R., 2002. Model Selection and Multimodel Inference: A Practical Information-Theoretic Approach, 2nd ed. Springer-Verlag, ISBN 0-387-95364-7.
Caffarra, A., Eccel, E., 2010. Increasing the robustness of phenological models for *Vitis vinifera* cv. Chardonnay. Int. J. Biometeorol. 54, 255–267.
Caffarra, A., Eccel, E., 2011. Projecting the impacts of climate change on the phenology of grapevine in a mountain area. Aust. J. Grape Wine Res. 17, 52–61.
Cannell, M.G.R., Smith, R.I., 1983. Thermal time, chill days and prediction of budburst in *Picea sitchensis*. J. Appl. Ecol. 20, 951–963.
Carbonell-Bejerano, P., Santa María, E., Torres-Pérez, R., Royo, C., Lijavetzky, D., Bravo, G., Aguirreolea, J., Sánchez-Díaz, M., Antolín, M.C., Martínez-Zapater, J.M., 2013. Thermotolerance responses in ripening berries of *Vitis vinifera* L. cv Muscat Hamburg. Plant Cell Physiol. 54, 1200–1216.
Castellarin, S.D., Matthews, M.A., Di Gaspero, G., Gambetta, G.A., 2007. Water deficits accelerate ripening and induce changes in gene expression regulating flavonoid biosynthesis in grape berries. Planta 227, 101–112.
Chuine, I., 2000. A unified model for budburst of trees. J. Theor. Biol. 207, 337–347.
Confalonieri, R., Acutis, M., Bellocchi, G., Donatelli, M., 2009. Multi-metric evaluation of the models WARM, CropSyst, and WOFOST for rice. Ecol. Model. 220, 1395–1410.
Corana, A., Marchesi, M., Martini, C., Ridella, S., 1987. Minimizing multimodal functions of continuous variables with the 'Simulated Annealing' algorithm. ACM Trans. Math. Softw. 13, 262–280.
Dokoozlian, N.K., 1999. Chilling temperature and duration interact on the budbreak of 'Perlette' grapevine cuttings. HortScience 34, 1–3.
Diodato, N., Bellocchi, G., 2007. Modelling solar radiation over complex terrains using monthly climatological data. Agric. Forest Meteorol. 144, 111–126.
Donatelli, M., Wösten, J.H.M., Bellocchi, G., 2005. Evaluation of pedotransfer functions. In: Pachepsky, Y.A., Rawls, W. (Eds.), Development of Pedotransfer Functions in Soil Hydrology. Developments in Soil Science, vol. 30. Elsevier, Amsterdam, pp. 357–363.
Duchêne, E., Huard, F., Dumas, V., Schneider, C., Merdinoglu, D., 2010. The challenge of adapting grapevine varieties to climate change. Climate Res. 41, 193–204.
Duchêne, E., Schneider, C., 2005. Grapevine and climatic changes: a glance at the situation in Alsace. Agron. Sustain Dev. 25, 93–99.
Fennel, A., Hoover, E., 1991. Photoperiod influences growth, bud dormancy and cold acclimation in *Vitis labruscana* and *Vitis riparia*. J. Am. Soc. Hort. Sci. 116, 270–273.
Fila, G., Di Lena, B., Gardiman, M., Storchi, P., Tomasi, D., Silvestroni, O., Pitacco, A., 2012. Calibration and validation of grapevine budburst models using growth-room experiments as datasource. Agric. Forest Meteorol. 160, 69–79.
Fishman, S., Erez, A., Couvillon, G.A., 1987. The temperature dependence of dormancy breaking in plants: mathematical analysis of a two-step model involving a cooperative transition. J. Theor. Biol. 124 (4), 473–483.

Friend, A.P., Trought, M.C., 2007. Delayed winter spur-pruning in New Zealand can alter yield components of Merlot grapevines. Aust. J. Grape Wine Res. 13, 157–164.
García de Cortázar-Atauri, I., Brisson, N., Gaudillere, J.P., 2009. Performance of several models for predicting budburst date of grapevine (*Vitis vinifera* L.). Int. J. Biometeorol. 53, 317–326.
García de Cortázar-Atauri, I., Chuine, I., Donatelli, M., Parker, A.K., Van Leeuwen, C., 2010. A curvilinear process-based phenological model to study impacts of climatic change on grapevine (Vitis vinifera L.) , XIth Congress of the European Society for Agronomy, August 29–September 3, Montpellier, France.
Goode, J., 2012. Viticulture: fruity with a hint of drought. Nature 492, 351–353.
Hlaszny, E., Hajdu, E., Bisztray, G.Y., Ladányi, M., 2011. Comparison of budburst models predictions for Kékfrankos. Appl. Ecol. Environ. Res. 10 (1), 75–86.
IPCC, 2007. Climate Change 2007: synthesis report. In: Pachauri, R.K., Reisinger, A. (Eds.), Contribution of Working Groups I, II and III to the Fourth Assessment Report of the Intergovernmental Panel on Climate Change. IPCC, Geneva, Switzerland.
Jackson, D.I., Lombard, P.B., 1993. Environmental and management practices affecting grape composition and wine quality: a review. Am. J. Enol. Vitic. 44, 409–430.
Jones, G.V., Davis, R.E., 2000. Climate influences on grapevine phenology, grape composition, and wine production and quality for Bordeaux, France. Am. J. Enol. Vitic. 51, 249–261.
Jones, G.V., Nelson, P., Snead, N., 2004. Modeling viticultural landscapes: a GIS analysis of the terroir potential in the Umpqua valley of Oregon. Geosci. Can. 31 (4), 167–178.
Jones, G.V., White, M.A., Cooper, O.R., Storchmann, K., 2005. Climate change and global wine quality. Clim. Change 73, 319–343.
Jones, G.V., Webb, L.B., 2010. Climate change, viticulture, and wine: challenges and opportunities. J. Wine Res. 21, 103–106.
Jones, N.K., 2012. The influence of recent climate change on wine regions in Quebec, Canada. J. Wine Res. 23, 103–113.
Laget, F., Tondut, J.-L., Deloire, A., Kelly, M.T., 2008. Climate trends in a specific Mediterranean viticultural area between 1950 and 2006. J. Int. Sci. Vigne. Vin. 42, 113–123.
Lavee, S., May, p., 1997. Dormancy of grapevine buds – facts and speculation. Aust. J. Grape Wine Res. 3, 31–46.
Linkosalo, T., Carter, T.R., Häkkinen, R., Hari, P., 2000. Predicting spring phenology and frost damage risk of *Betula* spp. under climatic warming: a comparison of two models. Tree Physiol. 20, 1175–1182.
Linkosalo, T., Lechowicz, M.J., 2006. Twilight far-red treatment advances leaf bud burst of silver birch (*Betula pendula*). Tree Physiol. 26, 1249–1256.
Linkosalo, T., Lappalainen, H.K., Hari, P., 2008. A comparison of phenological models of leaf bud burst and flowering of boreal trees using independent observations. Tree Physiol. 28, 1873–1882.
Lorenz, D.H., Eichhorn, K.W., Bleiholder, H., Klose, R., Meier, U., Weber, E., 1994. Phänologische Entwicklungsstadien der Weinrebe (*Vitis vinifera* L. ssp. vinifera). Viticult. Enol. Sci. 49, 66–70.
Kirkpatrick, S., Gelatt Jr., C.D., Vecchi, M.P., 1983. Optimization by simulated annealing. Science 220 (4598), 671–680.
Marsaglia, G., Zaman, A., Tsang, W.W., 1990. Toward a universal random number generator. Stat. Probabil. Lett. 8, 35–39.
Martin, S.R., Dunn, G.M., 2000. Effect of pruning time and hydrogen cyanamide on budburst and subsequent phenology of *Vitis vinifera* L. variety Cabernet Sauvignon in central Victoria. Aust. J. Grape Wine Res. 6, 31–39.
Menzel, A., 2005. A 500 year pheno-climatological view on the 2003 heatwave in Europe assessed by grape harvest dates. Meteorol. Z. 14, 75–77.
Mira de Orduña, R., 2010. Climate change associated effects on grape and wine quality and production. Food Res. Int. 43, 1844–1855.
Moncur, M.W., Rattigan, K., MacKenzie, D.H., Mc Intyre, G.N., 1989. Base temperatures for budbreak and leaf appearance of grapevines. Am. J. Enol. Vitic. 40, 21–26.
Mori, K., Goto-Yamamoto, N., Kitayama, M., Hashizume, K., 2007. Loss of anthocyanins in red-wine grape under high temperature. J. Exp. Bot. 58, 1935–1945.
Nash, J.E., Sutcliffe, J.V., 1970. River flow forecasting through conceptual models part I – a discussion of principles. J. Hydrol. 10, 282–290.
Neethling, E., Barbeau, G., Bonnefoy, C., Quénol, H., 2012. Change in climate and berry composition for grapevine varieties cultivated in the Loire Valley. Climate Res. 53, 89–101.
Nemani, R.R., White, M.A., Cayan, D.R., Jones, G.V., Running, S.W., Coughlan, J.C., Peterson, D.L., 2001. Asymmetric warming over coastal California and its impact on the premium wine industry. Climate Res. 19, 25–34.
Nendel, C., 2010. Grapevine bud break prediction for cool winter climates. Int. J. Biometeorol. 54, 231–241.
Oliveira, M., 1998. Calculation of budbreak and flowering base temperatures for *Vitis vinifera* cv. Touriga Francesa in the Douro region of Portugal. Am. J. Enol. Vitic. 49, 74–78.
Or, E., Nir, G., Vilozny., I., 1999. Timing of kydrogen cyanamide application to grapevine buds. Vitis 38, 1–6.
Parker, A.K., García de Cortázar-Atauri, I., Van Leewen, C., Chuine, I., 2011. General phenological model to characterise the timing of flowering and veraison of *Vitis vinifera* L. Aust. J. Grape Wine Res. 17, 206–216.
Petrie, P.R., Sadras, V.O., 2008. Advancement of grapevine maturity in Australia between 1993 and 2006: putative causes, magnitude of trends and viticultural consequences. Aust. J. Grape Wine Res. 14, 33–45.
Pope, V., Brown, S., Clark, R., Collins, M., Collins, W., Dearden, C., Gunson, J., Harris, G., Jones, C., Keen, A., Lowe, J., Ringer, M., Senior, C., Sitch, S., Webb, M., Woodward, S., 2007. The Met Office Hadley Centre climate modelling capability: the competing requirements for improved resolution, complexity and dealing with uncertainty. Philos. Trans. R. Soc. A 365 (1860), 2635–2657.
Pouget, R., 1964. Essai d'appreciation de l'intensité de la dormance chez quelques variétés et espèces de vigne. Vitis 4, 337–340.
R Development Core Team, 2012. R: A Language and Environment for Statistical Computing. R Foundation for Statistical Computing, Vienna, Austria, http://www.R-project.org/. (ISBN 3-900051-07-0).
Racsko, P., Szeidl, L., Semenov, M.A., 1991. A serial approach to local stochastic weather models. Ecol. Model. 57, 27–41.
Ramos, M.C., Jones, G.V., Martinez-Casasnovas, J.A., 2008. Structure and trends in climate parameters affecting winegrape production in northeast Spain. Climate Res. 38, 1–15.
Richardson, E.A., Seeley, S.D., Walker, D.R., 1974. A model for estimating the completion of rest for 'Redhaven' and 'Elberta' peach trees. HortScience 9, 331–332.
Rivington, M., Bellocchi, G., Matthews, K.B., Buchan, K., 2005. Evaluation of three model estimations of solar radiation at 24 UK stations. Agric. Forest Meteorol. 135, 228–243.
Sadras, V.O., Moran, M.A., 2012. Elevated temperature decouples anthocyanins and sugars in berries of Shiraz and Cabernet Franc. Aust. J. Grape Wine Res. 18, 115–122.
Schnabel, B.J., Wample, R.L., 1987. Dormancy and cold hardiness in *Vitis vinifera* L. cv White Riesling as influenced by photoperiod and temperature. Am. J. Enol. Vitic. 38, 265–272.
Schultz, H.R., Jones, G.V., 2010. Climate induced historic and future changes in viticulture. J. Wine Res. 21 (2-3), 137–145.
Semenov, M.A., Stratonovitch, P., 2010. Use of multi-model ensembles from global climate models for assessment of climate change impacts. Climate Res. 41, 1–14.
Sugeno, M., 1985. An introductory survey of fuzzy control. Inf. Sci. (NY) 36, 59–83.
Thompson, R., Clark, R.M., 2008. Is Spring starting earlier? Holocene 18, 95–104.
Valenti, L., Bravi, M., Dell'Orto, M., Ghiglieno, I., 2011. Influence of the winter pruning time on the phenology of some cultivars in the DOCG «Franciacorta» (Lombardy) and the DOC «Montefalco» (Umbria) , Proceedings of the 17th GiESCO Symposium, Asti-Alba (Italy) August 29th–September 2nd, pp. 339–342.
Vitasse, Y., François, C., Delpierre, N., Dufrêne, E., Kremer, A., Chuine, I., Delzon, S., 2011. Assessing the effects of climate change on the phenology of European temperate trees. Agric. Forest Meteorol. 151, 969–980.
Wang, J.Y., 1960. A critique of the heat unit approach to plant response studies. Ecology 41, 785–789.
Webb, L.B., Whetton, P.H., Barlow, E.W.R., 2007. Modelled impact of future climate change on the phenology of winegrapes in Australia. Aust. J. Grape Wine Res. 13, 165–175.
Williams, D.W., Andris, H.L., Beede, R.H., Luvisi, D.A., Norton, M.V.K., 1985. Validation of a model for the growth and development of the thompson seedless grapevine. II. Phenology. Am. J. Enol. Vitic. 36, 283–289.
Winkler, A., Cook, J.A., Kliewer, W.M., Lider, L.A., 1974. General Viticulture, 2nd revised edition. University of California Press, Berkeley and Los Angeles, California.
Wolfe, D.W., Schwartz, M.D., Lakso, A.N., Otsuki, Y., Pool, R.M., Shaulis, N.J., 2005. Climate change and shifts in spring phenology of three horticultural woody perennials in northeastern USA. Int. J. Biometeorol. 49, 303–309.
Yan, W., Hunt, L.A., 1999. An equation for modelling the temperature response of plants using only the cardinal temperatures. Ann. Bot. 84, 607–614.
Zadeh, L.A., 1965. Fuzzy sets. Inf. Control 8, 338–353.